INCH-POUND

MIL-C-70586B (AR)
10 October 1990
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CONTROL ASSEMBLY, M163A2 (SP): 9361150

This specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the control assembly used on Gun, Anti-Aircraft, Artillery, Self-Propelled 20MM, M163A2. The control assembly shall consist of a control panel and an elevation control subassembly. The control panel shall contain the controls and indicators required to operate the M163A2 weapon system. The control panel shall allow for the entry and display of ballistics and system fault information after static and dynamic checkout. The elevation control subassembly shall control the azimuth and elevation positioning of the gun and firing of the gun. Circuits in the control assembly shall be designed to enable target range and speed information to be set in manually by the gunner or automatically by the radar (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

MILITARY

MIL-F-13926 -	Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 -	Inspection Equipment, Acquisition, Main tenance and Disposition of
MIL-C-70575 -	Circuit Board, Display Control Panel (PIVADS): 9360884

STANDARDS

MILITARY

MIL-STD-109 -	Quality Assurance Terms and Definitions
MIL-STD-810 -	Environmental Test Methods and Engineering Guidelines
MIL-STD-2000-	Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

8437156 - Resistor Assembly
9361150 - Control Assembly, M163A2
9361217 - Remote BIT Tester
9361223 - RBT Test Cable

SPECIAL PACKAGING INSTRUCTION

PS9361150 - Packaging of Control Assembly, M163A2

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text

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of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9361150, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the control assembly shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The control assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
25 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 40 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The control assembly shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The control assembly shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

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3.3.3.2 Non-operating. The control assembly shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The control assembly shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 degrees F and +155 degrees F.

3.3.4.2 Non-operating (storage). The control assembly shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The control assembly shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the control assembly.

3.5 Performance.

3.5.1 Functional description. The control assembly shall consist of the control panel and the elevation control subassembly. The control assembly shall contain the controls and indicators required to operate the M163A2 weapon system, to display and enter ballistics and system fault information and to control the azimuth and elevation positioning and firing of the gun.

3.5.2 Performance characteristics.

3.5.2.1 Control panel. The control panel shall contain the following indicators and switches required to operate the weapon system in accordance with Drawing 9361150.

3.5.2.1.1 SYSTEM POWER switch (A1S5). An on/off, single pole/double throw toggle switch, with guard, to apply power to the system. With 24 V applied to J2-AW, SYSTEM POWER switch shall apply 24 Vdc to J2-AC.

3.5.2.1.2 SYSTEM POWER indicator (A1DS1). A lamp that indicates system power is on when lit. Indicator shall have a press-to-test capability.

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3.5.2.1.3 GUN POWER switch (A1S2). An on/off, single pole/double throw toggle switch, with guard, to apply power to the gun. With +24 V applied to J2-AK, the GUN POWER switch shall apply +24 V to J2-AH and J2-AX.

3.5.2.1.4 GUN POWER indicator (A1DS2). A lamp that indicates gun power is on when lit. Indicator shall have press-to-test capability.

3.5.2.1.5 COMPUTER POWER indicator (A1DS6). A lamp to indicate that power is available to the computer when lit. With +24 Vdc applied to J2-AU and ground applied to J3-K, the COMPUTER POWER indicator will illuminate. Indicator shall have press-to-test capability.

3.5.2.1.6 COMPUTER GOOD indicator (A1DS3). A lamp to indicate that the computer is operational when lit. With +24 Vdc applied to J2-AU and ground applied to J3-V, the COMPUTER GOOD indicator will illuminate. Indicator shall have press-to-test capability.

3.5.2.1.7 RADAR READY indicator (A1DS4). A lamp to indicate that the radar is operational when lit. With +24 Vdc applied to J3-X and ground applied to J2-BN, the RADAR READY indicator will illuminate. Indicator shall have press-to-test capability.

3.5.2.1.8 LOW VOLTAGE PROTECTION circuit (A1A5) and LOW VOLTAGE WARNING indicator (A1DS5). With +24 Vdc applied to J2-AW, continuity between J1-H and J1-J, and SYSTEM POWER switch A1S5 on, the LOW VOLTAGE PROTECTION circuit will apply +24 Vdc to J2-AC. When the continuity between J1-H and J1-J is removed, +24 Vdc will remain on J2-AC. When the voltage at J2-AW falls to 22.0-.22 Vdc, the LOW VOLTAGE PROTECTION circuit shall flash the LOW VOLTAGE WARNING indicator at approximately 1 Hz rate. When the voltage at J2-AW falls to 20.5 $\pm$ 0.21 Vdc, the LOW VOLTAGE PROTECTION circuit shall illuminate the LOW VOLTAGE WARNING indicator constantly and remove +24 Vdc from J2-AC. The LOW VOLTAGE PROTECTION circuit shall have an override capability that will allow the operator to prevent power shutdown, or to restart the system after shutdown has occurred. The LOW VOLTAGE PROTECTION circuit shall have a self test capability as follows. When self test is initiated, the LOW VOLTAGE PROTECTION circuit shall flash the LOW VOLTAGE WARNING indicator for 5.5 $\pm$ 2.5 s after which it will illuminate steadily and +24 Vdc will be removed from J2-AC. Following self test, +24 Vdc shall be restored to J2-AC by normal startup procedure. The LOW VOLTAGE WARNING indicator shall have press-to-test capability.

3.5.2.1.9 MODE switch (A1S1). A switch with five settings to initiate the following. With +24 V applied to J2-AU, MODE switch shall apply +24 Vdc to the pins indicated for each position and continuity between pins as indicated:

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GRD	Ground mode	J3-p	---
RADAR	Radar mode	J3-b, J2-AS, J2-BE	---
MAN	Manual mode	J3-E, J2-BE	J3-h and J3-S
EXT	External mode	J3-m	J3-h and J2-AP
TEST	Test mode	J3-d, J2-BE	J3-h and J3-S

3.5.2.1.10 LOW VOLTAGE switch (A1S8). A double pole/double throw center off toggle switch to initiate the following:

TEST — Self test of LOW VOLTAGE PROTECTION circuit as described in 3.5.2.1.8. This shall be a momentary contact position.

OVERRIDE — Override operation of the LOW VOLTAGE PROTECTION circuit as described in 3.5.2.1.8.

NORMAL — Normal operation of the LOW VOLTAGE PROTECTION circuit as described in 3.5.2.1.8.

3.5.2.1.11 FAULT indicator (A1DS7). A lamp to indicate a system fault when lit. With +24 Vdc applied to J2-AW, system power switch AIS5 set to on, and ground applied to J3-e, the FAULT indicator will illuminate. Indicator shall have press-to-test capability.

3.5.2.1.12 FAULT BYPASS switch (A1S13). A double pole/double throw toggle switch, with guard, to bypass the ACTION SWITCH LOCK-OUT relay and allow continued system operation. With +24 Vdc applied to J2-AW, the SYSTEM POWER switch on, and ground applied to J3-e,the FAULT BYPASS switch shall apply +24 Vdc to J3-Z. In addition, it shall override the function of the ACTION SWITCH LOCKOUT relay (3.5.2.1.19) and re-establish continuity of the ACTION switch circuits.

3.5.2.1.13 FIRING RATE switch (A1S4). A 4 pole/5 position switch to set the firing rate, with the following switch settings: HI BURST LIMIT (10, 30, 60, 100) and LO-NO LIMIT. The FIRING RATE switch consists of two decks, each deck consisting of two sections (total of four sections). The fourth section of the FIRING RATE switch selects one of five burst limit resistive networks. The burst limit resistive networks shall present a different resistance for each position of the FIRING RATE switch between pins J2-r and J2-n. The resistance shall be adjustable between the limits shown in figure 1 for each switch position.

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SWITCH POSITION	RESISTANCE AT -40 DEGREES F		RESISTANCE AT AMBIENT		RESISTANCE AT AMBIENT	
	AIR5-AIR9 SET FULL CCW		AIR5-AIR9 SET FULL CCW		AIR5-AIR9 SET FULL CW	AIR5-AIR9 SET FULL CCW
LO-NO LIMIT	6.82K-	7.32K	715.53-	1215.4	23.22K- 29.04K	235.23- 735.23
HI RATE						
10RDS	40.35K-	43.24K	33.59K-	37.11K	123.59K- 145.11K	33.10K- 35.77K
30RDS	48.86K-	54.19K	33.63K-	39.30K	213.63K- 255.30K	33.11K- 37.78K
60RDS	65.80K-	77.44K	34.97K-	48.92K	484.97K- 588.92K	33.41K- 44.08K
100RDS	79.96K-	92.21K	35.00K-	49.19K	485.00K- 589.19K	33.41K- 44.08K

FIGURE 1. Switch Positions

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In addition, the FIRING RATE switch shall provide continuity between pins as follows:

SWITCH POSITION	CONTINUITY
LO-NO LIMIT	J2-p and J2-q J2-p and J2-j J2-k and J2-m
HI RATE-any burst	J2-i and J2-j

3.5.2.1.14 GUN CLEAR switch (A1S3). A double pole/double throw toggle switch with three settings to initiate the following:

OFF	Gun does not clear after firing.
AUTO	Automatically clear the gun of used shells.
OPERATOR	Manually clear the gun of used shells.

With +24 V applied to J2-AK, the GUN CLEAR switch shall provide continuity and +24 V output at the following pins:

POSITION	CONTINUITY	+24 Vdc
OFF	None	None
AUTO	J2-x and J2-y	J2-w
OPERATOR	J2-x and J2-z	J2-AE

3.5.2.1.15 ROUNDS REMAINING indicator (A1M1). A digital readout to show the status of ammunition. For each capacitor discharge pulse applied to J2-AY, the ROUNDS REMAINING indicator shall decrement the tens digit by one count. The ones digit shall be fixed at zero. The indicator shall be manually settable by the operator.

3.5.2.1.16 RANGE potentiometer (A1R3). A variable potentiometer, graduated in meters, to set target distance. The control shall provide a variable voltage from 0 to +15 Vdc at J3-h when +15 V is applied to J3-f and when the MODE switch is in the MAN or TEST position.

3.5.2.1.17 TARGET SPEED potentiometer (A1R4). A variable potentiometer, graduated in knots, to set target speed. With +15 V applied to J3-f, the control shall provide a variable voltage from 0 to +15 V at J3-T.

3.5.2.1.18 SIGHT LAMP potentiometer (A1R1). A variable potentiometer to adjust sight reticle brightness. With the system power switch on the control shall provide a variable resistance between pin J2-AW and pins J1-D and J1-F from 10 to 70 ohms.

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3.5.2.1.19 Action switch lockout relay (A1K1). A double pole/double throw relay which shall inhibit gun/turret motion under fault conditions with a ground applied to J3-a, +24 V applied to J2-AW, and system power on the relay. The lockout relay shall interrupt the action switch circuit to prevent gun/turret motion.

3.5.2.2 BALLISTICS/BIT circuit. The BALLISTICS/BIT circuit shall contain the following indicators and switches necessary to allow the operator to communicate with the fire control processor, display BIT messages, and enter Ballistics and BIT data in accordance with Drawing 9361150.

3.5.2.2.1 FUNCTION switch (A1S14). A single-pole, 5-deck rotary switch with twelve settings to initiate the following:

The switch shall provide continuity between connector P2 pins as follows:

POSITION	FUNCTION	CONTINUITY WITH P2-51	CONTINUITY WITH P2-59
OFF	Ballistics/BIT off	---	P2-18, P2-14, P2-15, P2-16, P2-17
LT	Lamp test	P2-14	P2-43, P2-15, P2-16, P2-17
MV	Muzzle velocity	P2-15	P2-14, P2-16, P2-17
AD	Air density	P2-14, P2-15	P2-16, P2-17
AT	Ammo type	P2-16	P2-14, P2-15, P2-17
WS	Wind speed	P2-14, P2-16	P2-15, P2-17
WD	Wind direction	P2-15, P2-16	P2-14, P2-17
T	Tilt	P2-14, P2-15, P2-16	P2-17
CT	Cross tilt	P2-17	P2-14, P2-15, P2-16
B	BIT	P2-14, P2-17	P2-15, P2-16
ZO	Zero order	P2-15, P2-17	P2-14, P2-16
SP	Spare	P2-14, P2-15, P2-17	P2-16

3.5.2.2.2 DISPLAY/ENTRY switch (A1S9). A single pole/single throw toggle switch to allow display or entry of ballistics and system fault information. In the ENTRY position, the switch shall provide continuity between P2-10 and P2-59.

3.5.2.2.3 VALUE display (A1DS8). A 4 digit digital readout to display ballistic and system fault information.

3.5.2.2.4 SELECT pushbutton (A1S10). A single pole/single throw pushbutton switch to select a VALUE display digit for updating. When depressed, the switch shall provide continuity between P2-11 and P2-59.

3.5.2.2.5 INCREMENT pushbutton (A1S11). A single pole/single throw pushbutton switch to change the selected VALUE display digit. When depressed, the switch shall provide continuity between P2-12 and P2-59.

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3.5.2.2.6 ENTER pushbutton (A1S12). A single pole/single throw pushbutton switch to store the selected ballistic values or to initiate system fault testing. When depressed, the switch shall provide continuity between P2-13 and P2-59.

3.5.2.3 Elevation control subassembly. The elevation control subassembly shall contain the following indexes and switches in accordance with Drawing 9361150.

3.5.2.3.1 Trigger (A4S2). A momentary pushbutton to fire the gun. When depressed, the switch shall provide continuity between J2-AF and J2-AJ.

3.5.2.3.2 Action switches (A3S1 and A4S1). Switches allow elevation and azimuth control. When either switch is depressed, system power switch is on and +24V is applied to J2-AU and J2-AW, +24V shall be provided to J3-A. With a 100 ohm load connected to J2-AD, it shall also provide +24V to J2-AD when:

(1) The Mode Switch is in either Ground or Test.

(2) The Mode Switch is in either Radar, Manual or External, and only after the Acquisition/Track switch has been momentarily depressed.

3.5.2.3.3 ACQ/TRK switch (A3S2). A momentary contact closure switch to indicate acquisition or track. With +24 V applied to J2-AU, the switch shall provide +24 V to J3-s when depressed.

3.5.2.3.4 Elevation hand control (A2R1). A potentiometer to control elevation positioning of the gun. The potentiometer shall be in accordance with Drawing 8437156. With +15 V applied to J3-f and -15 V applied to J3-M, the control shall provide 0 to $\pm$ 9.5 V minimum between J3-D and J3-n.

3.5.2.3.5 Azimuth hand control (A1R10). A potentiometer to control azimuth positioning of the gun. The potentiometer shall be in accordance with Drawing 8437156. With +15 V applied to J3-f and -15 V applied to J3-M, the control shall provide 0 to $\pm$ 9.5 V minimum between J3-P and J3-F.

3.5.3 Display control BIT assembly. The control assembly shall contain a 4-layer circuit board, mounted horizontally within the control panel. The BIT assembly shall interface with the BALLISTICS/BIT circuit to transmit switch (A1S9-A1S14) status to the fire control processor, conduct end-around test and load alphanumeric data for display on the VALUE display. The BIT assembly performance shall be in accordance with MIL-C-70575.

3.6 Test points. The control assembly shall contain test points as specified in Table I (Connector J5 test point pin designations).

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TABLE I. Connector J5 test point pin designations.

PIN	FUNCTION
A	SIGHT LAMP control test point
b	FIRING RATE switch test point
B	MAN MODE switch position test point
c	FIRING RATE switch test point
C	MODE switch test point
E	External range enable test point
G	ACQ/TRK switch test point
J	EXT MODE switch position test point
R	ROUNDS REMAINING test point
S	MODE switch test point
T	RADAR power test point
U	Action switch test point
V	Trigger switch test point
W	Trigger switch test point
X	SYSTEM POWER test point
Y	RANGE potentiometer test point
Z	TARGET SPEED potentiometer test point

3.7 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.8 Soldering. The control assembly shall be soldered per the general requirements and Task F of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set

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forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable documents. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 degrees to +90 degrees F.
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

4.3 First article inspection.

4.3.1 First article submission. The requirement for first article and the responsibility (Government or contractor) for the first article testing shall be as specified in the contract and the applicable drawing. The sample for the first article tests shall consist of three control assemblies manufactured in the same manner, using the same materials, equipment, processes, and procedure as in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

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4.3.1 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II (Functional Test) and Table III (Special Sampling) herein and to any or all requirements of the applicable drawings.

4.3.1.1 Government testing. When the Government is responsible for conducting first article tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to ensure that it conforms to all the requirements of the contract and the applicable drawings. The contractor shall submit a record of this inspection with the sample including certified test report (CTR) or certificate of conformance (COC) for the materials.

4.3.1.2 Contractor testing. When the Contractor is responsible for conducting first article tests, the sample shall be inspected by the contractor for all the requirements of the contract, all of the examinations and tests specified in Table II (Functional Test) and Table III (Special Sampling) herein and to all requirements of the applicable drawings. The sample and a record of this inspection, including CTR/COC, shall be submitted for the Government approval. The Government reserves the right to witness the Contractor's inspection.

4.3.2 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

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4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One control assembly out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

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TABLE III. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed tolerance limit

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and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV (Test Equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list.

ITEM NUMBER		DESCRIPTION
1.	Power supply	±24 Vdc + 1 V, 1A.
2.	Power supply	±15 Vdc ± 0.25 V, 1A.
3.	Digital voltmeter (DMV)	± 30 Vdc range, ± 0.1 Vdc accuracy, 20 Kohms input impedance.
4.	Ohmmeter	1 megohm range, 1% accuracy.
5.	Remote BIT Tester (RBT)	TS-4211/TSM-100B, Drawing 9361217.
6.	RBT test cable	Drawing 9361223.
7.	Load Resistor	100 ohm, 10 watts.
8.	Resistor R1	50 ohm, 5 watts.
9.	Capacitor C1	1100 Mf +5%-10%, 40 Vdc.
10.	Switch S1	Single pole/double throw switch.

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

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4.7.1.1 Vibration. The test shall be performed as specified in Method 514.3, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The control assemblies shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the control assemblies shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516.3 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the control assemblies shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500.2 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502.2 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507.2 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

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4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance.

4.7.3.1 Control panel. The control panel shall be tested to meet the requirements of 3.5.2.1.

4.7.3.1.1 SYSTEM POWER switch. The SYSTEM POWER switch shall be tested to meet the requirements of 3.5.2.1.1.

4.7.3.1.2 SYSTEM POWER indicator. The SYSTEM POWER indicator shall be tested to meet the requirements of 3.5.2.1.2.

4.7.3.1.3 GUN POWER switch. The GUN POWER switch shall be tested to meet the requirements of 3.5.2.1.3.

4.7.3.1.4 GUN POWER indicator. The GUN POWER indicator shall be tested to meet the requirements of 3.5.2.1.4.

4.7.3.1.5 COMPUTER POWER indicator. The COMPUTER POWER indicator shall be tested to meet the requirements of 3.5.2.1.5.

4.7.3.1.6 COMPUTER GOOD indicator. The COMPUTER GOOD indicator shall be tested to meet the requirements of 3.5.2.1.6.

4.7.3.1.7 RADAR READY indicator. The RADAR READY indicator shall be tested to meet the requirements of 3.5.2.1.7.

4.7.3.1.8 LOW VOLTAGE indicator. The LOW VOLTAGE indicator shall be tested to meet the requirements of 3.5.2.1.8.

4.7.3.1.9 MODE switch. The MODE switch shall be tested to meet the requirements of 3.5.2.1.9.

4.7.3.1.10 LOW VOLTAGE switch. The LOW VOLTAGE switch shall be tested to meet the requirements of 3.5.2.1.10.

4.7.3.1.11 FAULT indicator. The FAULT indicator shall be tested to meet the requirements of 3.5.2.1.11.

4.7.3.1.12 FAULT BYPASS switch. The FAULT bypass switch shall be tested to meet the requirements of 3.5.2.1.12.

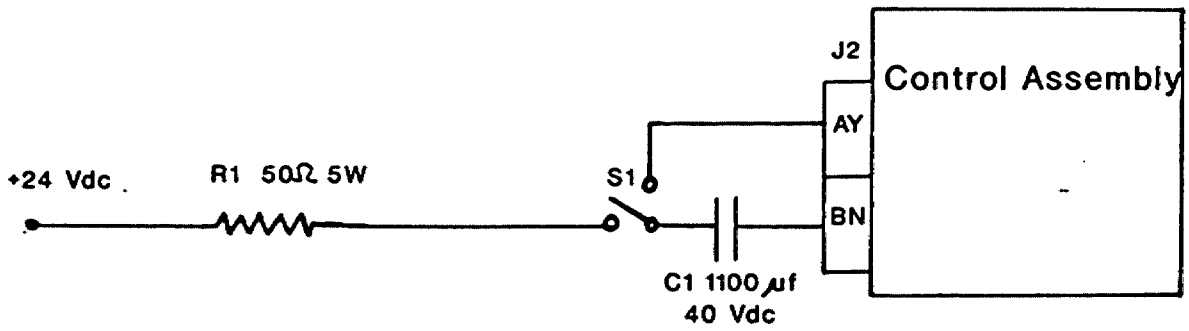
4.7.3.1.13 FIRING RATE switch. The FIRING RATE switch shall be tested to meet the requirements of 3.5.2.1.13.

4.7.3.1.14 GUN CLEAR switch. The GUN CLEAR switch shall be tested to meet the requirements of 3.5.2.1.14.

4.7.3.1.15 ROUNDS REMAINING indicator. The ROUNDS REMAINING indicator shall be tested, per the following test set-up, to meet the requirements of 3.5.2.1.15:

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TEST SETUP



4.7.3.1.16 RANGE potentiometer. The RANGE potentiometer shall be tested to meet the requirements of 3.5.2.1.16.

4.7.3.1.17 TARGET SPEED potentiometer. The TARGET SPEED potentiometer shall be tested to meet the requirements of 3.5.2.1.17.

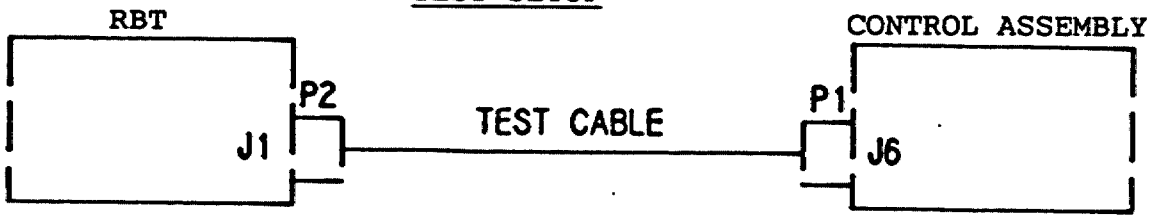
4.7.3.1.18 SIGHT LAMP potentiometer. The SIGHT LAMP potentiometer shall be tested to meet the requirements of 3.5.2.1.18.

4.7.3.1.19 Action switch lockout relay. The action switch lockout relay shall be tested to meet the requirements of 3.5.2.1.19.

4.7.3.2 BALLISTICS/BIT circuit. The BALLISTICS/BIT circuit shall be tested to ensure that indication of the presence and nature of detected faults shall be provided to inform the operator of system GO/NO-GO status and to allow for further troubleshooting and fault isolation. The BALLISTICS/BIT circuit shall be tested to ensure that the operator has the capability to enter and change BIT and ballistics data on the VALUE display, command display lamps tests and turn off the display. The BALLISTICS/BIT circuit shall be tested to meet the requirements of 3.5.2.2 in accordance with the following procedures:

a. Connect the control assembly to the Remote BIT Tester (RBT) as per the following test setup:

TEST SETUP



b. Set the RBT controls as follows:

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SWITCH	POSITION
STATION 1234	0 0 0 0
SEND	DOWN
UNIT 4 TEST/NORMAL TEST	NORMAL TEST

c. Set RBT power ON/OFF pushbutton switch to ON.

d. Set Ballistics BIT Panel FUNCTION switch to LT. The VALUE display LEDs shall light to 8 8 8 8.

e. Set Ballistics BIT Panel FUNCTION switch to OFF. The VALUE display LEDs shall extinguish.

f. Set RBT SEND switch to the up position. Set RBT STATION 1234 switch to the following sequences as per Table V (RBT STATION 1234 switch settings). The Ballistics BIT VALUE display shall respond with the indicated symbols in the non-X positions. Symbols appearing in the X position are not critical and may vary.

TABLE V. RBT STATION 1234 switch settings.

STATION 1234	VALUE DISPLAY
1 0 1 0	XXX0
1 1 1 1	XXX1
1 2 1 2	XXX2
1 3 1 3	XXX3
1 4 1 4	XXX4
1 5 1 5	XXX5
1 6 1 6	XXX6
1 7 1 7	XXX7
1 8 1 8	XXX8
1 9 1 9	XXX9
3 0 3 0	XX0X
3 1 3 1	XX1X
3 2 3 2	XX2X
3 3 3 3	XX3X
3 4 3 4	XX4X
3 5 3 5	XX5X
3 6 3 6	XX6X
3 7 3 7	XX7X
3 8 3 8	XX8X
3 9 3 9	XX9X
7 0 7 0	X0XX
7 1 7 1	X1XX
7 2 7 2	X2XX
7 3 7 3	X3XX
7 4 7 4	X4XX
7 5 7 5	X5XX
7 6 7 6	X6XX
7 7 7 7	X7XX
7 8 7 8	X8XX
7 9 7 9	X9XX

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TABLE V (continued).

STATION 1234	VALUE DISPLAY
9 0 9 0	0XXX
9 1 9 1	1XXX
9 2 9 2	2XXX
9 3 9 3	3XXX
9 4 9 4	4XXX
9 5 9 5	5XXX
9 6 9 6	6XXX
9 7 9 7	7XXX
9 8 9 8	8XXX
9 9 9 9	9XXX
9 10 9 10	AXXX
9 11 9 11	CXXX
9 12 9 12	EXXX
9 13 9 13	FXXX
9 14 9 14	-XXX
9 15 9 15	UXXX
8 1 8 1	=XXX
8 2 8 2	=XXX
8 3 8 3	=XXX
8 4 8 4	=XXX
8 5 8 5	=XXX
8 6 8 6	=XXX
8 7 8 7	=XXX
8 8 8 8	=XXX
8 9 8 9	=XXX
8 10 8 10	=XXX
8 11 8 11	=XXX
8 12 8 12	=XXX
8 13 8 13	=XXX
8 14 8 14	=XXX
8 15 8 15	=XXX

g. Set RBT STATION 1234 switch as follows. The RBT BIT CHANNEL 1 indicator shall light as appropriate. NOTE: ON = X, OFF = 0.

	CHANNEL 1							
STATION 1234	DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
5 5 5 5	0	X	0	X	0	X	0	X
10 10 10 10	X	0	X	0	X	0	X	0

h. Set STATION 1234 switch to 12 0 12 0.

i. Set Ballistics BIT FUNCTION switch to OFF.

j. Set Ballistics BIT DISPLAY/ENTER switch to DISPLAY. The RBT CHANNEL 1 BIT indicator shall indicate as follows:

DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
X	X	X	X	0	0	0	0

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k. Set DISPLAY/ENTER switch to ENTER. The RBT CHANNEL 1 BIT indicators shall indicate as follows:

DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
0	X	X	X	0	0	0	0

l. Press ENTER switch on Ballistics BIT panel. The RBT CHANNEL 1 BIT indicators shall indicate as follows:

DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
0	0	X	X	0	0	0	0

m. Press INCREMENT switch on Ballistics BIT panel. The RBT CHANNEL 1 BIT indicators shall indicate as follows:

DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
0	X	0	X	0	0	0	0

n. Press SELECT switch on Ballistics BIT panel. The RBT CHANNEL 1 BIT indicators shall indicate as follows:

DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
0	X	X	0	0	0	0	0

o. Set Ballistics BIT FUNCTION switch to the following positions as per Table VI (Ballistic BIT FUNCTION switch settings). The CHANNEL 1 BIT indicators shall indicate as per Table VI.

TABLE VI. Ballistic BIT FUNCTION switch settings.

FUNCTION SWITCH POSITION	CHANNEL 1 DS1	DS2	DS3	DS4	DS5	DS6	DS7	DS8
LT	0	X	X	X	0	0	0	X
MV	0	X	X	X	0	0	X	0
AD	0	X	X	X	0	0	X	X
AT	0	X	X	X	0	0	0	0
WS	0	X	X	X	0	X	0	X
WD	0	X	X	X	0	X	X	0
T	0	X	X	X	0	X	X	0
CT	0	X	X	X	X	0	0	0
B	0	X	X	X	X	0	0	X
ZO	0	X	X	X	X	0	X	0
SP	0	X	X	X	X	0	X	X

4.7.3.2.1 ENTER pushbutton. The ENTER pushbutton shall be tested to meet the requirements of 3.5.2.2.6.

4.7.3.3 Elevation control subassembly. The elevation control subassembly shall be tested to meet the requirements of 3.5.2.3.

4.7.3.3.1 Trigger. The trigger shall be tested to meet the requirements of 3.5.2.3.1.

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4.7.3.3.2 Action switch. The action switch shall be tested to meet the requirements of 3.5.2.3.2.

4.7.3.3.3 ACQ/TRK switch. The ACQ/TRK switch shall be tested to meet the requirements of 3.5.2.3.3.

4.7.3.3.4 Elevation hand control. The elevation hand control shall be tested to meet the requirements of 3.5.2.3.4.

4.7.3.3.5 Azimuth hand control. The azimuth hand control shall be tested to meet the requirements of 3.5.2.3.5.

4.7.3.4 Display control BIT assembly. The BIT assembly shall have met the requirements of MIL-C-70575.

4.7.4 Test points. The control assembly shall be tested to meet the requirements of 3.6.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 to be specified in 3.8.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instructions PS9361150.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The control assembly is used on the M163A2 and consists of the controls and indicators necessary to operate the M163A2 weapon system.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.
f. Certificate of conformance for each lot or shipment of product.

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Anti-aircraft artillery
Elevation control subassembly
Fire Control System
Gun, anti-aircraft artillery
Self-Propelled 20MM M163A2

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publication and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1285-A747)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-70586B	2. DOCUMENT DATE (YYMMDD) 10 October 1992

3. DOCUMENT TITLE CONTROL ASSEMBLY, M163A2 (SP): 9361150

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets if needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S ARMY ARDEC STANDARDIZATION OFFICE	b. TELEPHONE (Include Area Code) (1) Commercial (201) 724-6674 (2) AUTOVON 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMACA-BAC-S PICATINNY ARSENAL, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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